Exercice n°3

1- Déterminer c et φ graphiquement ?

* cohésion :

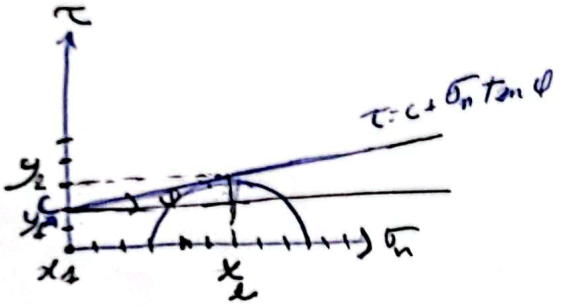
est l'intersection de droite de rupture et l'axe ordonnées.

⇒ A partir de graphe :

$c = 20\ KPa$

* Angle de frottement :

est la pente de droite de rupture. (Droite)



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$\tan\varphi = \frac{y_2 - y_1}{x_2 - x_1} = \frac{61 - 20}{120 - 0}$

$\tan\varphi = \frac{41}{120} \Rightarrow \varphi = 18,86°$

2) On suppose que : $\varphi = 20° / c = 20° / \sigma_3 = 73\ KPa$

calculer contrainte principale majeur σ_1 ?

On a :

$(\sigma_1 + \sigma_3)\sin\varphi - (\sigma_1 - \sigma_3) + 2c\cdot\cos\varphi = 0$ ... (1)

On développe l'équation (1) :

$(\sigma_1 + \sigma_3)\sin\varphi = (\sigma_1 - \sigma_3) + 2c\cos\varphi = 0$

$\sigma_1\sin\varphi + \sigma_3\sin\varphi - \sigma_1 + \sigma_3 + 2c\cdot\cos\varphi = 0$

$\sigma_1(\sin\varphi - 1) + \sigma_3(\sin\varphi + 1) + 2c\cdot\cos\varphi = 0$

$\sigma_1(\sin\varphi - 1) = -\sigma_3(\sin\varphi + 1) - 2c\cos\varphi$

$\sigma_1 = \frac{-\sigma_3(\sin\varphi + 1) - 2c\cdot\cos\varphi}{\sin\varphi - 1}$

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$\sigma_1 = \frac{-[\sigma_3(\sin\varphi + 1) + 2c\cdot\cos\varphi]}{-(\sin\varphi + 1)}$

$\sigma_1 = \frac{\sigma_3(\sin\varphi + 1) + 2c\cdot\cos\varphi}{1 - \sin\varphi}$ ... (2)

on peut utilis (1) ou (2) pour calculer σ_1 :

$\sigma_1 = 208\ KPa$

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3- Déterminer σ_n et τ suivant R :

$$\begin{cases} \sigma_n = \left(\frac{\sigma_1 + \sigma_3}{2} + \frac{\sigma_1 - \sigma_3}{2}\cos 2\theta\right) \\ \tau = \frac{\sigma_1 - \sigma_3}{2}\sin 2\theta \\ \theta = \frac{\pi}{4} + \frac{\varphi}{2} \Rightarrow \theta = 110° \end{cases}$$

$\sigma_n = 116,83\ KPa$

$\tau = 62,67\ KPa$

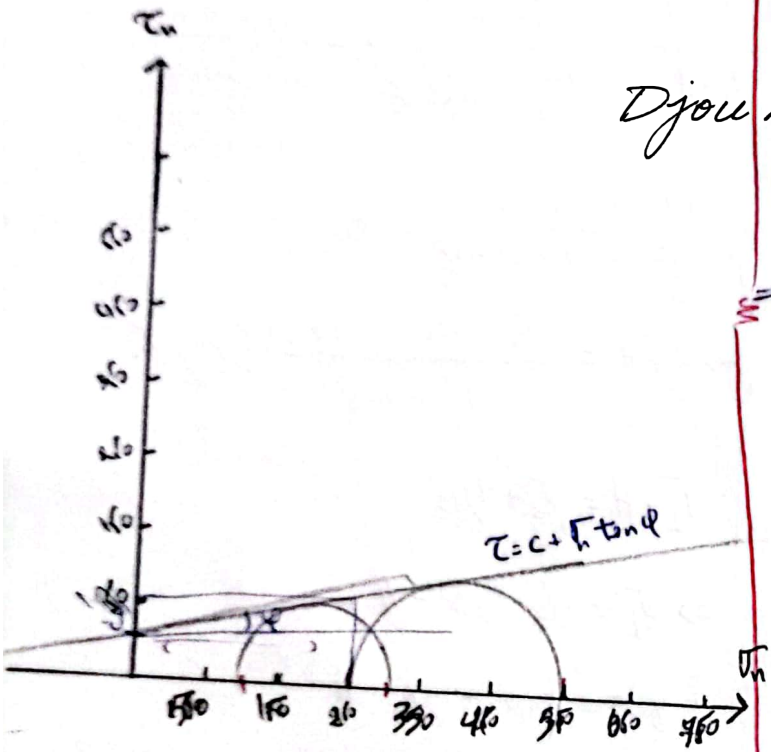
Exercice n°8

Essai 1 : $\sigma_3 = 100$ Kpa / $\sigma_1 = 300$ Kpa

Essai 2 : $\sigma_3 = 250$ Kpa / $\sigma_1 = 550$ Kpa

1- tracer les cercles de Mohr.

2- déduire c et φ graphiquement.



~~c=0~~

$c = 58$ Kpa

b) $\tan\varphi = \frac{y}{x} = \frac{7.2}{36.0}$

$\varphi = 11,62°$

3- déduire c et φ analytiquement :

On a :

$(\sigma_1 + \sigma_3)\sin\varphi - (\sigma_1 - \sigma_3) + 2c\cos\varphi = 0$

il y a deux essais :

$$\begin{cases} 400\sin\varphi - 200 + 2c\cos\varphi = 0 \quad \cdots (1) \\ 800\sin\varphi - 300 + 2c\cos\varphi = 0 \quad \cdots (2) \end{cases}$$

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(1) - (2) $\Rightarrow 400\sin\varphi + 100 = 0$

$\Rightarrow 400\sin\varphi = 100$

$\sin\varphi = \frac{100}{400} \Rightarrow \boxed{\varphi = 14,03°}$

en remplace : - - (1)

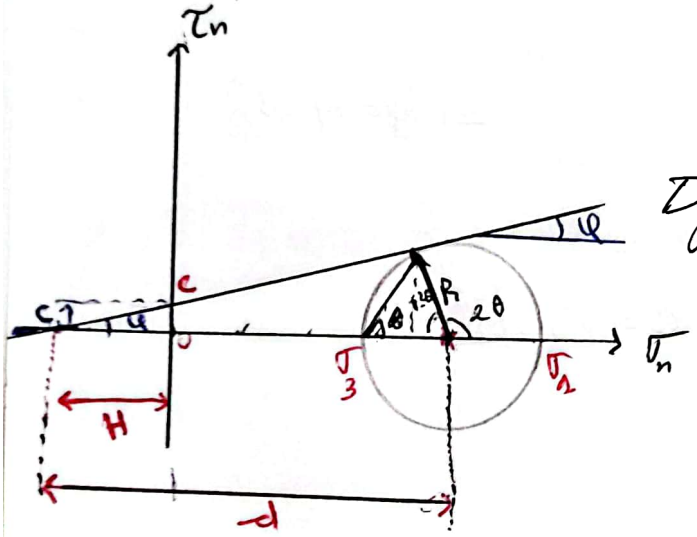
$c = 53,08$ Kpa

Exercice (Thabet) :

On a élaboré deux essais triaxiaux sur deux échantillons dont les résultats suivant :

1ère ESS : $\sigma_1 = 800$ KPA $\quad \sigma_3 = 200$ KPA.

2ème ESS : $\sigma_1 = 1100$ KPA $\quad \sigma_3 = 300$ KPA.



$$\begin{cases} \sin\varphi = \dfrac{R}{d} \Rightarrow d = \dfrac{R}{\sin\varphi} \\ \tan\varphi = \dfrac{c}{H} \Rightarrow H = \dfrac{c}{\tan\varphi} \end{cases}$$

$$\begin{cases} \sigma_1 = d - H + R \Rightarrow \sigma_1 + H = d + R \\ \sigma_3 = d - H - R \Rightarrow \sigma_3 + H = d - R \end{cases}$$

$$\sigma_1 + H = d + R \Rightarrow \sigma_1 + H = \frac{R}{\sin\varphi} + R$$

$$\Rightarrow \sigma_1 + H = \frac{R}{\sin\varphi} + \frac{R(\sin\varphi)}{\sin\varphi}$$

$$= \frac{R + R(\sin\varphi)}{\sin\varphi}$$

$$\sigma_1 + H = R \cdot \frac{(1+\sin\varphi)}{\sin\varphi}$$

Alors :

$$\sigma_3 + H = d - R$$

$$= \frac{R}{\sin\varphi} - R$$

$$\sigma_3 + H = \frac{R(1-\sin\varphi)}{\sin\varphi}$$

$$\frac{\sigma_1 + H}{\sigma_3 + H} = \frac{\frac{R(1+\sin\varphi)}{\sin\varphi}}{R\left(\frac{1-\sin\varphi}{\sin\varphi}\right)} = \frac{1+\sin\varphi}{1-\sin\varphi}$$

$$\frac{\sigma_1 + H}{\sigma_3 + H} = \frac{1+\sin\varphi}{1-\sin\varphi} = K$$

$$\Rightarrow \sigma_1 + H = \frac{(\sigma_3 + H)(1+\sin\varphi)}{1-\sin\varphi} = (\sigma_3 + H)K$$

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$$\sigma_1 + H = (\sigma_3 + H)K$$

$$\Rightarrow \sigma_1 = (\sigma_3 + H)K - H$$

$$\begin{cases} 800 = (200 + H)K - H \\ 1100 = (300 + H)K - H \end{cases}$$

$$\Rightarrow \begin{cases} 800 = 200K + H(K-1) \quad \cdots (2) \\ 1100 = 300K + H(K-1) \quad \cdots (1) \end{cases}$$

$(1) - (2) \Rightarrow 300 = 100K$

$$\boxed{K = 3}$$

$$\frac{1+\sin\varphi}{1-\sin\varphi} = 3 \Rightarrow 1 + \sin\varphi = 3(1 - \sin\varphi)$$

$$\Rightarrow 1 + \sin\varphi = 3 - 3\sin\varphi$$

$$\Rightarrow 4\sin\varphi = 2$$

$$\Rightarrow \sin\varphi = \frac{1}{2}$$

$$\boxed{\varphi = 30°}$$

Suite (Exo Thabet) :

$$\begin{cases} 800 = 200K + H(K-1) \\ 1100 = 300K + H(K-1) \end{cases}$$

$K = 3$ / $\varphi = 30°$

$800 = 600 + 2H$

$200 = 2H$

$\boxed{H = 100}$

Ona :

$\tan\varphi = \frac{c}{H} \Rightarrow c = \tan\varphi \cdot H$

$= \tan 30° \times 100$

$\boxed{c = 57{,}74 \text{ KPa}}$

Deuxième Méthode :

~~$(\sigma_1 + \sigma_3) - (\sigma_1 - \sigma_3)$~~

$(\sigma_1 + \sigma_3)\sin\varphi - (\sigma_1 - \sigma_3) + 2c\cos\varphi = 0$

$$\begin{cases} (800 + 200)\sin\varphi - (800 - 200) + 2c\cos\varphi = 0 \\ (1100 + 300)\sin\varphi - (1100 - 300) + 2c\cos\varphi = 0 \end{cases}$$

$$\begin{cases} 1000\sin\varphi - 600 + 2c\cos\varphi = 0 \quad \cdots ② \\ 1400\sin\varphi - 800 + 2c\cos\varphi = 0 \quad \cdots ① \end{cases}$$

① − ② $\Rightarrow 400\sin\varphi - 200 = 0$

$\Rightarrow \sin\varphi = \frac{1}{2}$

$\boxed{\varphi = 30°}$

Dans ① :

$1400\sin 30 - 800 + 2c\cos 30° = 0$

$1400\sin 30 + 2c\cos 30° = 800$

$2c\cos 30° = 800 - 1400\sin 30°$

$2c = \frac{800 - 1400\sin 30°}{\cos 30°}$

$c = \frac{800 - 1400\sin 30°}{2\cos 30°}$

$\boxed{c = 57{,}74 \text{ KPA}}$

Djoc Az

(Concours d'accès à l'école Exogénéraux)

2	$\gamma = 19$	$K_0 = 0,7$
4m	$\gamma = 17$	$K_0 = 0,5$
2m	$\gamma = 20$	$K_0 = 0,6$

σ_v | u | σ'_v | σ'_h | σ_h en fonction de profondeur z.

$\sigma_v = \gamma . z$

$\sigma_{v_1} = 19 . z \Rightarrow 38$ kPa

$\sigma_{v_2} = 17 z + 4 \Rightarrow 106$ kPa

$\sigma_{v_3} = 20z + 14 \Rightarrow 146$ kPa

$u = \gamma_w . h_w = \gamma_w . z = 10 . z$

$u_1 = 10 . z$ / $u_2 = 10 . z$ / $u_3 = 10 . z$

$\sigma'_v = \sigma_v - u$

$\sigma'_{v_1} = 19z - 10z \Rightarrow \sigma'_{v_1} = 9z$

$\sigma'_{v_2} = 17z + 4 - 10z = \sigma'_{v_2} = 7z + 4$

$\sigma'_{v_3} = 10z + 14$

$K = \frac{\sigma'_h}{\sigma'_v} \Rightarrow \sigma'_h = K_0 . \sigma'_v$

$\sigma'_{h_1} = K_0 . 9z \Rightarrow 0,7 \times 9z$

$\sigma'_{h_1} = 6,3z$

$\sigma'_{h_2} = 3,5z + 2$

$\sigma'_{h_3} = 10z - 2,4$

Djou Az